

Yara Elshehawi

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🔗 My Portfolio Website

Profile

Digital Media Engineering graduate from the German University in Cairo, with a strong focus on AI, computer vision, and deep learning. I have hands-on experience in machine learning through academic projects and internships. I'm currently looking for opportunities in AI/ML where I can contribute and continue growing and I'm available to start immediately.

EDUCATION

IGCSE Graduate 2020

MET Student in German University in Cairo 2020 – 2025

EXPERIENCE

Machine Learning Intern, GUC 07/2023 – 08/2023

- Contributed to a research lab project focused on developing algorithms for weapons detection and age/gender identification.

Junior Teaching Assistant, GUC 10/2023 – 12/2023

Database I

- Contributed in teaching database concepts to undergraduates

NOTABLE PROJECTS

Bachelor Project — Adapting a Human-Agent Toolkit to a Physical Robot Head 📄

- Used MediaPipe's Blendshape model with Python and NVIDIA Omniverse to track and extract facial expressions in real time
- Mapped blendshape coefficients from webcam input to robotic actuators to achieve synchronized, human-like lip sync and emotional expressions

Medical Image Segmentation using Deep Learning 📄

- Trained a U-Net model on brain MRI scans to detect tumor regions
- Used OpenCV for preprocessing; evaluated using Dice Coefficient and IoU
- Deployed the model in Hugging Face Space

Motion Analysis using MEI & MHI 📄

- Built motion history/energy representations from video sequences to track human activity
- Useful for recognizing motion direction and intensity in real time

Dynamic Video Summarization 📄

- Tracked and overlaid moving objects on reference frames using OpenCV and Euclidean distance tracking
- Created condensed visual summaries of traffic footage with timestamps

Foreground Detection and Image Binarization using Classical CV 📄

- Separated moving foreground from video using a running average background model
- Binarized galaxy images via unsupervised EM-based pixel classification.

Interactive Frequency-Domain Image Processing Tool

- Built and deployed an interactive FFT-based image processing tool that visualizes frequency spectra and applies real-time low-pass, high-pass, and band-pass filters using NumPy, OpenCV, and Gradio (hosted on Hugging Face Spaces)

LANGUAGES

Arabic — Native

English — Fluent

German — Basic

TECHNICAL SKILLS

Languages

Python, Java, C++(academic), SQL, JavaScript

Libraries/Tools

TensorFlow, Docker(basic), OpenCV, Mediapipe, Hugging Face Agents

EXTRACURRICULARS

MUN (Model United Nations) — Started as a delegate, progressing to board member for two years, ultimately serving as The Presidency of the International Court of Justice (ICJ). Gained strong leadership, public speaking, and organizational skills, moderating court proceedings and managing complex discussions, while deepening knowledge of diplomacy and international relations.

IEEE — Maintained team morale and organization while attending technical sessions to broaden my knowledge of engineering and AI.

Student Curriculum Committee, GUC — Improved student experience by providing feedback on courses and collaborating with faculty.

Awards

DAAD Extra Scholarship Award, DAAD

Awarded to students with outstanding academic performance.

Bachelor Thesis Scholarship, German University in Cairo

Awarded to the top-ranked student for exceptional academic performance to complete my Bachelor thesis in Germany.

Courses

Fundamentals of Agents, Hugging Face

2025

Learned how AI agents use LLMs to reason and act autonomously; built my first functional AI agent with Hugging Face tools.